Chapter

Personalized Learning with AI: Adapting Education to Learners' Needs

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Abstract

This chapter explores the transformative role of artificial intelligence (AI) in enabling personalized learning (PL). It draws on the theoretical foundations of constructivism, multiple intelligences, self-determination theory, connectivism, and data-driven decision-making to examine how AI supports learner engagement, autonomy, and self-regulation. It highlights intelligent tutoring systems, recommender engines, emotion-aware tools, and real-time feedback systems, which furnish relevant and adaptable learning pathways. Global case studies – such as Summit Public Schools, Carnegie Learning, Squirrel AI, and David Game College – are discussed to illustrate the effectiveness of these data-driven, human-AI collaboration-based, and scalable adaptive personalized instructional systems. The chapter also addresses ethical, pedagogical, and human-related implications and emphasizes the challenges related to algorithm bias, data privacy, and teacher readiness. Emerging trends, like the integration of AI with extended reality (XR) environments that aim toward a more immersive, experiential, and evidence-based learning, are also discussed.

Keywords: artificial intelligence in education, personalized learning, adaptive learning systems, learning analytics, educational technology, edtech

1. Introduction

Personalized learning is founded on the premise that learning is a unique experience for learners and must be tailored to their individual needs, focusing on their learning pace and progression [1]. Though delivering personalized learning has been on the agenda of educationists since the eighteenth century, as seen from early models like the monitorial system and Keller Plan, the lack of practical resources to implement such learning has prevented its implementation [2].

However, with the advancement of digital technologies and artificial intelligence (AI), these limitations are largely overcome. Personalized learning is now in a new phase of transition, where customized and tailored instructions and content can be accessed by learners at the time and location that are most convenient to them. Adaptive learning systems based on AI technology enable the analysis of students' data in real time, which leads to the adjustment of content in a dynamic manner to offer customized learning pathways [3]. AI-supported learning platforms are therefore affecting how learners perceive their self-efficacy, engage with content, and progress at their own pace with confidence. Such AI-integrated learning environments are flexible and encourage content mastery in a dynamic environment, which is often not possible within the rigid classroom structures [4].

This chapter explores how AI-powered systems are transforming personalized learning and also addresses the challenges and ethical dilemmas that have emerged due to AI-personalized learning integrations. It is expected to provide a foundational and critical understanding of the opportunities and limitations of using AI in personalized learning environments.

The specific objectives of this chapter are:

1. To identify the fundamental factors contributing to building the AI-supported personalization frameworks.

2. To explore how AI technologies enable key functions of personalized learning.

3. To examine the pedagogical, ethical, and equity implications of AI-driven personalized learning.

4. To illustrate real-world implementation models of AI-supported personalized learning in diverse educational contexts.

5. To determine the key success factors in scaling AI-driven personalized learning solutions through international case studies.

One of the strategies adopted in AI-integrated personalized learning includes the development of comprehensive learner profiles, which guide the creation of learner-focused content customization and encourage the development of competency-based assessments. Secondly, predictive analytics is employed to develop AI-enhanced assessment tools that enable early intervention and tailored support in real time, thus leading to better academic outcomes for students [5]. Additionally, the in-built flexibility in the learning environment enables learners to pace their education and take ownership, thus providing both motivation and autonomy [6]. Educators, too, can obtain actionable insights and provide feedback in real time.

These strategic elements shape the fundamentals of building AI-personalized learning frameworks, as seen in Objective 1. By emphasizing intentional design focused on data analytics, learner profiling, and content adaptability, these elements enable scalable customization. Objective 2 further highlights the functions of personalized learning as rooted in real-time feedback, adaptive progression, and predictive assessment to deliver learner-centered education.

However, some critical pedagogical, ethical, and systemic challenges and implications are highlighted in Objective 3. A major concern is related to issues like data privacy, algorithmic bias, and teacher marginalization, which may act as barriers to the adoption of AI-based learning personalization. Data privacy is a global concern, and ethical frameworks are often fragmented or not supported by adequate transparency at the institutional level, which may lead to both ethical and legal repercussions [7]. Additionally, algorithmic bias is a major concern, as most AI-based platforms and applications are trained on segregated data, which may lead to bias or discrimination against diverse users. Further, AI systems do not offer a similar human connection, intuition, or ethical judgment that educators can bring to the students [8, 9]. The impacts of such ethical and decision-making aspects of personalized AI learning require further exploration and research, which could help establish ethical guidelines and ensure teacher readiness through professional development.

This chapter also addresses real-world examples and implementation models (Objective 4) and discusses international case studies that underscore the significance of local contexts and multistakeholder involvement. By drawing upon diverse educational settings like the charter schools in the USA, the private institutions in the UK, and learning centers in China, these case studies provide insights into the real-world impact made by AI-supported personalized learning. The case studies highlight contextual factors such as culture, policy making, and diverse pedagogical approaches on the success of models like Squirrel AI, David Game College, and the Summit Public Schools (SPS) model.

Finally, Objective 5 leads us to identify key success factors based on the insights from the practical applications of AI in personalized learning. These key factors are gleaned from the above case studies and underline the importance of the robustness of data analytics, the feedback-loop-based nature of personalized learning, collaborative environments, and teachers' training. In addition to technology and training, successful implementation is dependent on inclusive and localized educational policies and support from top management.

2. Fundamental factors in building an AI-enhanced personalized learning

2.1 What is personalized learning?

Personalized learning, though not a new concept, is being revolutionized by advancements in AI [4]. This section explores the fundamental concepts and definitions of AI in personalized learning and discusses its significant and transformative potential. Personalized learning is both an educational approach and a practical strategy that enables a customized learning experience for students [5]. It aims to develop and deliver instruction to individual students based on their unique characteristics, learning strengths, requirements, and interests [7]. Personalized learning is, therefore, facilitated by flexible learning environments, student ownership of learning, and ongoing assessment and dynamic adjustment of learning strategies. Several theories underpin personalized learning and its advantages.

2.2 Theoretical underpinnings of personalized learning

2.2.1 Constructivist theory

2.2.1.1 Overview

Constructivist theory [10, 11] suggests that knowledge is constructed through active participation and interaction of learners with their environments and social context [12]. Students are actively engaged in their learning process and indulge in meaning-making from experiences. Piaget suggests that there are developmental stages where learners interact with objects and internalize abstract thinking [10]. His theory is largely applied to the learning stages in early childhood; a similar logic is present in the learning-by-doing approach in higher education. Learning is made effective not only by personal and intimate interaction with content but also through social mediation, where peers and instructors play a major role [11].

Personalized learning is also supported through understanding social connections and knowledge sharing, which impact learners' learning pathways [13]. Vorobyeva et al. postulate that platforms offering adaptive feedback and learner-controlled pacing are well-aligned with constructivist pedagogy [14]. Similarly, Holmes et al. suggest that AI tools extend constructivist learning environments by encouraging learners in iterative learning and meaning-making [15].

2.2.1.2 Application in personalized learning

Research has indicated that constructivist environments supported by AI generative and adaptive technologies lead to an enhancement in students' engagement [16]. Guo et al. support integrating generative AI into constructivist contexts, as AI-generated simulations and peer collaboration prompts were found to be linked to increases in students' motivation, creativity, and collaboration [17]. By developing curricula and activities that allow students to explore, ask questions, and build upon prior knowledge, educationists can encourage highly personalized learning.

Personalized learning is further enhanced through project-based learning and student-directed projects, as done at SPS in California, where constructivist principles are applied to enhance student engagement, autonomy, and academic growth [18]. Summit's Personalized Learning Plans (PLPs) help students set weekly goals and track their progress. The PLP platform enables self-reflection and guidance by mentors, which creates a sense of ownership for their learning [19]. By encouraging students to inquire and engage in problem-solving to draw their own conclusions, teachers can engage in constructive learning and facilitate personalization. This is also reflected in project-based learning, where students learn by doing and report a deeper understanding of the knowledge, as well as better satisfaction with the learning process [20].

Pham et al.'s constructivist-based system using Google Codelabs resulted in higher satisfaction levels, with a mean score of 3.33–3.44 on a 4-point scale for system quality, content alignment, and relevance [11]. Similarly, Do et al. report that within three months of implementing constructionist environments via the design science approach, 193 project management students showed improvement in their motivation and learning strategy scores, with intrinsic motivation significantly predicting learning strategies ($\beta = 0.292–0.315$, $p < 0.001$) and extrinsic motivation also showing strong effects ($\beta = 0.277–0.322$, $p < 0.001$) [21]. Additionally, Marienko et al. have shown that adaptive AR systems improved perceived usability and learner satisfaction in experiential training contexts and simulation labs, which was found to translate into reported improved actual task performance [22].

2.2.2 Multiple intelligences theory

2.2.2.1 Overview

Howard Gardner's multiple intelligences theory presumes that intelligence is diverse and can be differentiated in the form of linguistic, logical-mathematical, spatial, musical, interpersonal, intrapersonal, bodily-kinesthetic, and naturalistic [23]. These multiple intelligences vary among different learners and need to be considered and challenged when designing curriculum instruction. A recent study using 149 teachers showed a significant positive correlation ($r = 0.511$, $p < 0.001$) between teachers' use of technology to support multiple intelligences and student engagement [24]. Additionally, research has found students improve by 25% on metrics like satisfaction with curriculum and confidence in learning after changes were made to their curricula to focus on multiple intelligences [25].

2.2.2.2 Application in personalized learning

While this approach is still in its nascent stages, there is some evidence suggesting that curricula or instructional designs aligning with students' more prominent intelligences are likely to lead to improved performance and engagement [13]. For example, by using a fuzzy-AHP-based e-content sequencing system that matched content with students' main intelligence, it was possible to instill 84.6% higher engagement and interest among students compared to those learning with traditional approaches [26]. In a study conducted across 41 public schools in the USA, Kornhaber et al. found that 78% of schools that adopted multiple intelligence frameworks for three years were able to report higher test scores, better student behavior, substantial gains for students with learning disabilities, and enhanced parental participation [27]. Instructional pluralism, which is at the core of personalized learning, enables learners to approach problems using their specific cognitive strengths, facilitating deeper learning [28]. Moreover, it has also been found that integrating multiple intelligences theory with educational technology is increasingly being deployed in inclusive and diverse learning settings [29].

2.2.3 Self-determination theory

2.2.3.1 Overview

Self-determination theory (SDT), developed by Deci and Ryan, is based on the understanding that learners have basic psychological needs in the form of the need for autonomy, the need to feel competent, and the need to relate to the content [30]. The drive to fulfill these needs acts as an intrinsic motivation for learners [6]. Deci and Ryan emphasize that when educational environments are structured to support these needs, students display deeper learning and psychological well-being. In contrast, in learning settings where there is a lack of autonomy and high pressure to perform, students are likely to feel disengaged [30]. Other studies have also validated the link between autonomy-supportive environments and enhanced learning, engagement, and overall improvement in student well-being [6, 31].

2.2.3.2 Application in personalized learning

By allowing students to feel autonomy and control, personalized learning using AI technology applies principles of SDT to improve student motivation and engagement [32]. Research has found that mobile learning systems designed using SDT principles showed a 16% improvement in intrinsic motivation for learning and a 12% improvement in learning outcomes compared to students using traditional platforms [33]. Additionally, meaningful and relevant personalized content enhances engagement levels and develops students' competence and confidence levels [34]. SDT theory suggests that intrinsic motivations ensure that learning through constant engagement with content improves students' resilience and self-efficacy in the longer term [32]. Personalized online learning that encourages students to participate autonomously has been shown to significantly boost students' perceived autonomy, competence, and motivation [1].

2.2.4 Connectivism

2.2.4.1 Overview

George Siemens and Stephen Downes' connectivism theory suggests that learning predominantly occurs through connections and networks [35]. This framework has particular relevance in the digital age, as social media platforms and the internet allow learners to access vast amounts of information and collaborate with others globally. Furthermore, with AI-linked knowledge management platforms, such linkages and connections can be more targeted and relevant to student learning. Connectivism theory also suggests that building connections not only improves access to information and knowledge but also helps learners contribute their knowledge, which is an essential aspect of learning in a knowledge economy [36]. Connectivism directly extends to the current AI-supported learning environments that provide adaptive and flexible knowledge ecosystems for learners to keep abreast of new information and knowledge aligned with their requirements [37, 38].

2.2.4.2 Application in personalized learning

This theory applies directly to personalized learning environments, where students can interact with the content, peers, and teachers through online learning platforms that enable the creation of both social and academic connections and provide incentives through recognition and appreciation for knowledge sharing and engagement. A US Department of Education report recently indicated that schools integrating online collaboration tools into their instructional designs see substantial improvements in students' critical thinking and communication skills [39]. Additionally, other studies have shown that students' engagement in networking activities over digital platforms leads to higher self-efficacy when the interactions are aligned with personal learning goals [40].

For example, a study using students in a connectivism-driven collaborative online international learning (COIL) highlighted connectivism's role in enhancing understanding and application of complex concepts like diversity and inclusion, and showcased students' significant ($p < 0.05$) increase in willingness to apply knowledge as compared to a control group [20].

A significant improvement in engagement scores and substantially improved motivation levels has been reported by using ViLLE (a collaborative learning platform developed by the University of Turku, Finland) compared to traditional instruction [41]. Similarly, the Student Learning Space (SLS), by the Ministry of Education in Singapore, provides AI-based recommendations and collaboration tools for students, which are linked to improved content retention and classroom participation [42]. Further, Hardof-Jaffe and Peled reported a connection between connectivist-based teacher training and six key growth areas, including breaking time/place boundaries, enhanced personal and professional development, and digital peer interaction [43]. These growth areas, in turn, are related to teachers' ability to develop and support personalized and adaptive learning environments.

2.2.5 Data-driven decision making

2.2.5.1 Overview

This framework is central to many AI-based personalized learning systems and forms the basis of most personalized learning platforms [44].

Recent studies have shown how data-driven instructions can enable real-time detection of learning gaps and generate appropriate proactive, rather than reactive, responses from teachers [45]. This proactive approach fosters a personalized learning pathway for the learner [46].

2.2.5.2 Application in personalized learning

Data-driven personalized learning platforms such as DreamBox Learning, Knewton Alta, Squirrel AI, and CENTURY Tech provide customized learning pathways, monitor engagement, and enable real-time assessment [47–50]. By leveraging data analytics to customize instruction, schools can demonstrate improved learning gains at a rate 1.5 times the national average, as found in a review of personalized learning models funded by the Bill & Melinda Gates Foundation [44]. When analytical tools are integrated into learning platforms, students can receive alerts, personalized feedback, and motivational messages that enable them to engage more with their learning [51]. Data-driven learning systems like DreamBox Learning identify learning gaps in specific demographic sections and help deliver personalized and customized learning solutions [52].

Other platforms, like Knewton Alta, deliver STEM content based on a system of continuously updating learner profiles in real time [49]. Similarly, Squirrel AI in China uses deep learning to provide microlevel personalization for students and enables real-time feedback based on adaptive instructional design [50, 53].

CENTURY Tech (a UK-based platform utilizing machine learning applications that help generate dynamic learner profiles and diagnostics for students and teachers) reduces teacher planning time by 30% and increases student content mastery by up to 20% over a term across secondary schools in the UK [47].

3. Key dimensions of AI-supported personalized learning

3.1 Supporting learner engagement and motivation through AI

AI enhances personalized learning through several mechanisms, predominantly by focusing on a learning-centric model that keeps the focus on the learning abilities, motivations, intelligence type, and autonomy of the learner [4]. This is achieved by accessing and analyzing historical usage data to predict outcomes and deliver appropriate interventions to students using predictive analytics. AI models use natural language processing (NLP) to facilitate learners' interactive experiences through chatbots that provide on-demand assistance or questions that align with the learners' current learning. Additionally, AI-underpinned content recommendations customize the learning path using an algorithm to guide decision-making, as seen in Khan Academy learning systems. These AI tools not only deliver personalized learning but also provide feedback and present choices to learners, which enhances their sense of autonomy and competency and keeps them motivated [54]. By designing learning modules supported by theories like SDT, delivering intrinsic motivations, and providing appreciation, points, or token rewards, these AI-enabled learning tools enhance students' learning by utilizing their psychological inclinations. Additionally, using predictive analytics, it is possible to assess any increasing frustration levels, which can then trigger the system to adapt to less challenging tasks, thus enabling the students to continue learning [55].

3.2 Personalization through learner-centric instructional design

A key concept in personalized learning is the adoption of learner-centric models underpinned by theories of constructionism, which focus on students' interests and learning styles, enabling greater engagement [56, 57]. Several studies have already linked personalized learning environments to higher engagement and retention rates. For example, a study by the RAND Corporation reported that students who engaged in personalized learning gained approximately three percentile points in mathematics and four percentile points in reading compared to those who engaged in traditional learning [58]. Personalization is made possible by aligning pedagogy with individual learner needs and preferences and encouraging their ownership and choice. Learner profiling tools that are available on emerging digital platforms enable this integration, help in identifying the students' strengths, goals, and interests, and assist in relevant content delivery and assessment [59].

3.3 Adaptive systems for real-time instructional support

The key focus of personalized learning is to continually adapt to a learner's performance, offer differentiated support, and foster confidence and academic growth [60]. Adaptive learning technologies enable the customization of learning by collecting, analyzing, and using students' learning activity and interaction with material to develop an understanding of their preferences, strengths, and patterns [61]. Using these insights, tailored learning pathways are generated to address students' learning gaps in real-time, enabling targeted and focused learning with an adequate challenge level for students, keeping them motivated without frustration [62, 63]. More recent developments in adaptive systems encourage learner modeling, where AI can construct a student's profile and provide affective support to encourage self-reflection and goal setting [64]. For instance, DreamBox Learning utilizes an adaptive math program for K–8 students, and it has been found that students who used DreamBox for at least 14 hours showed significant improvement in math proficiency [48].

3.4 Using data-driven insights for instructional decision making

Personalized learning using AI is based on data-driven instructional strategies. Not only are students provided with customized education, but even educators can obtain insights about students' learning trajectories using multiple reports generated by AI from assessments, engagement metrics, and behavioral patterns. These insights help identify if a student is struggling, disengaging, or performing well, enabling timely interventions and tailored support. In some cases, the data-driven instructional design and assessment insights are found to lead to a 5% increase in graduation rates at Western Governors University in Utah [65]. Moreover, such systems track progress over time and can showcase long-term trends that can potentially be used for improving curricula and planning instructions. By employing formative analytics, teachers can adapt their instructions in real time based on data collected from students [46]. Based on the performance level as well as learning preferences, students' learning can be adjusted accordingly, and teachers can proactively prevent disengagement and frustration among students [45, 66].

3.5 Enhancing self-regulated learning through feedback mechanisms

Continuous feedback is a critical component of personalized learning. With AI systems, students can obtain instantaneous feedback and make amendments. AI-powered feedback systems, such as intelligent tutoring systems (ITS) and automated writing evaluators, support learning and encourage independent learning habits where students can align their goals, stay motivated, and improve by following the feedback received [67]. According to a study published in Frontiers in Psychology, feedback has a significant positive effect on student learning outcomes, reinforcement of correct understanding, and self-improvement [68]. Moreover, students are likely to develop a habit of revision based on feedback. In addition, the autogenerated nature of feedback reduces reliance on delayed teacher evaluation and facilitates autonomy and engagement. Self-regulation is fostered by these systems and empowers students to actively participate in goal setting, monitoring their strategies, and pacing themselves [4]. It also strengthens learned metacognition by helping them understand and reflect on their errors, identify strengths, and move toward the next levels of learning.

4. AI technologies for personalized learning

4.1 Intelligent tutoring systems (ITS)

AI-driven ITS emulate human tutors by displaying the ability to guide and adapt. By taking into consideration the real-time responses of the students and identifying their cognitive patterns, these systems tailor instruction suited to the learner. ITS also delivers personalized feedback and learning material, which enables personalized learning solutions. For example, the Korbit platform, which is used for data science studies, uses ML and NLP to deliver adaptive tutorials and has been found to lead to learning gains of 2–2.5 times higher than those of students in traditional MOOCs [66]. ITS stimulates an interactive tutoring experience to foster a sense of autonomy and paced progression, which are essential for personalized learning.

Carnegie Learning's MATHia platform is another ITS that has been implemented across several US school districts. In a randomized controlled study, it was found that middle school students using MATHia achieved gains two to three times higher than those students working with standard instruction [69]. The real-time adaptation and data-rich learning experience enable personalized learning at scale [70].

4.2 Recommender systems for learning pathways

Recommender systems in education are used to suggest learning pathways and resources that are aligned with the learner's performance and take into account their preferences and educational goals. Recommender systems use ML techniques like content-based algorithms and collaborative filtering, which create tailored learning pathways for students and lead to high recall rates, accurate understanding, enhanced engagement, and course completion [71]. Coursera data suggests that their AI-driven recommender engine, which provided personalized suggestions based on past behavior and user profiles, resulted in an increase in performance, achieving precision: 0.98%, recall: 1.00, F1 score: ≈0.99%, and accuracy: 0.98% [72]. Coursera's recommender system analyzes users' behaviors and performance, and detects skill gaps and goals to align with suitable recommendations. The recommender systems typically use dynamic profiling of users, which is based on behavior and performance data, to assess needs across diverse segments of users [73].

4.3 Emotion-aware and affective computing tools

Affective computing technologies (ACT) are built to detect and respond to learners' emotional states while delivering customized instructions. Adapting content delivery is made possible by the inclusion of biometric sensors or physical analytics based on facial recognition or the assessment of voice modulation, and it leads to learner engagement and well-being. ACTs use convolutional neural networks that enable the detection of learners' affective states like confusion, interest, or boredom, which then trigger suitable adaptive pedagogical responses like a change in the pace of instructions, reduction in cognitive overload, and delivery of motivational prompts [74]. The real-time emotion-based feedback provided by affective systems has been found to increase students' persistence by over 30%, especially among students who have prior records of exhibiting low performance [75].

4.4 Real-time assessment and feedback mechanisms

Real-time assessment and feedback mechanics evaluate user performance continuously and provide immediate feedback and adjustment of instructions, enabling early intervention. Platforms like Amira Learning assess students' oral reading skills and provide instant personalized help. Amira's learning analytics data indicate that students using the platform improve in proficiency over a short period of time, with significant benefits for learners with dyslexia [76]. Similarly, students using the Eedi platform in the UK, which uses real-time diagnosis and AI analytics to provide instant feedback to learners, experience improved math concept retention by 22% over a period of five weeks [77].

AI technologies are transforming personalized learning by delivering adaptive, data-driven, and emotion-informed learning experiences. ITS can emulate human tutors, while recommender systems, like those used by Coursera, empower users to fill their skill gaps. Affective computing tools recognize emotional states and enable resilience and persistence in learning. Similarly, real-time assessment platforms deliver instant feedback, enabling early intervention and improvement.

5. Implications of AI-driven personalized learning

5.1 Ethical and equity considerations in AI-powered personalization

With the rapid evolution and deployment of AI in education, there are growing concerns about the ethical usage of learner data and algorithm bias that could lead to educational disparities. If AI systems are trained on biased or incomplete datasets, there is an increased risk of discrimination against certain societal sections, leading to the reinforcement of existing educational discrimination [70, 78]. Additionally, as most AI-led educational platforms operate within what is called a black-box due to the lack of understanding of their functionality, there is an inherent lack of transparency in algorithm-led decision-making. This may lead to potential confusion or a lack of acceptance regarding the way content is presented or paced for certain learners. As such, integrations of AI in learning modules need to be audited for bias and must follow ethical and regulatory frameworks. In recent years, there has been a growing deployment of explainable AI, or XAI, which allows for more transparency in content recommendations and shows learners why certain pathways were opened for them [55].

5.2 Human-AI collaboration in personalized learning environments

With all the benefits of AI in the personalization of learning, the human element continues to be critical. Teachers are required to interpret and make sense of the AI-generated insights, especially as learners may be subjected to algorithmic bias and may not obtain the correct information. Teachers are needed for contextual learning within the academic, social, or cultural frameworks [7]. The role of a teacher as a mentor offering knowledge and insights, but also emotional support and motivation, cannot be understated. As such, AI-led personalization needs to take into account the expertise of teachers and should not be viewed as a replacement but rather as an augmentation of that expertise. By enabling teachers to provide insights and context within the framework of personalized learning, it is possible to blend automation and personalization and deliver better value to learners.

5.3 Pedagogical implications of AI-based personalized learning

The integration of AI in personalized learning leads to a pedagogical shift, reframing the roles of teachers, learners, and instructional strategies. One key implication is the transition from content delivery to facilitation. With AI systems offering real-time feedback, content recommendations, and adaptive sequencing, educators are increasingly positioned as orchestrators of learning rather than traditional teachers. Teachers must now engage in dynamic instructional decision-making, often in response to algorithmic insights [14]. For instance, AI-powered dashboards allow educators to identify disengagement or learning gaps early, enabling timely scaffolding or intervention. This shifts the pedagogical emphasis from reactive remediation to proactive guidance. Moreover, the use of AI tools enables teachers to interact with both formative and summative analytics for grading, feedback, and lesson planning. However, these changes also present challenges. If AI recommendations are treated as authoritative rather than contextual, it may lead to incoherence. Teachers must be trained in using AI tools but also learn to critically evaluate their output [13]. Finally, algorithms may not fully account for linguistic, cultural, or neurodiverse learning preferences unless guided by a teacher's contextual understanding.

6. Implementation in practice: Global case studies

6.1 Case study 1: Summit Public Schools – California's personalized learning pioneer

SPS is a network of charter schools spanning the states of Washington and California that introduced a PLP, which is delivered through a platform that tracks students' learning goals, performance, and assessments [18]. There are separate dashboards for students, parents, and teachers, which allow for shared accountability and a culture of transparency. Teachers at SPS are able to use data analytics to assess students' progress and customize content that targets the development of students' cognitive skills and understanding of concepts, rather than just the completion of the syllabus or the attainment of grades [18]. The platform also has provisions for mentors who can review and guide the students. By placing the student at the center and allowing them to set their own learning goals, select assignments that they are interested in, and learn at a pace that is comfortable for them, PLP is one of the leading programs in personalized learning [79]. Summit's model has been found to lead to improvements in critical thinking skills, problem-solving skills, and student engagement and motivation [69].

6.2 Case study 2: Carnegie learning – AI-powered math instruction

Carnegie Learning's AI-driven Cognitive Tutor for Algebra I is an ITS that uses students' responses to diagnose and explain misconceptions, dynamically changing their task sequencing and difficulty levels. It mirrors expert tutors, follows their cognitive processes, and has been found to improve mathematics performance by eight percentile points for high school students. It has also been recognized by the US Department of Education for meeting educational standards [80]. Students in the second year, however, seemed to have shown more significant gains (an eight percentile increase) than in the first year [59].

Additionally, Carnegie Learning's MATHia platform (available for middle and high school students in several US school districts) is based on a composite metric, Adaptive Personalized Learning Score (APLSE), that tracks students' mastery level, effort, and time-on-task. It was found from school data from Illinois, Ohio, South Carolina, and Washington State that a 10-point increase in APLSE score results in a 0.13–0.55 standard deviation boost in standardized math tests, or about 4–17 point gains in standard assessments [81, 82]. Furthermore, an increase of 400% in passing rates in math exams was observed over three years across Brockton, Massachusetts, districts using MATHia [83].

6.3 Case study 3: Squirrel AI – personalized learning at scale in China

Squirrel AI is a Chinese edtech company that uses an adaptive learning platform based on a knowledge graph system [84]. Squirrel AI breaks down subjects into microconcepts for easy understanding. It uses dynamic profiling of learners by collecting data from millions of data points and predicts knowledge gaps that can be filled with personalized instructions and content. Squirrel AI is unlike teacher-led teaching, as it uses an automated system to assess, teach, and track progress. Currently, the system is adopted in over 3,000 learning centers across China and serves 24 million students [85]. Research has indicated tangible benefits of Squirrel AI over traditional classroom-based instruction [50, 53]. The company claims nearly five to ten times improvement in learning efficiency over traditional methods [85].

6.4 Case study 4: David game college – Chatgpt and AI tutoring in the UK

In a first-of-its-kind model, David Game College, London, introduced a "teacherless" classroom for 20 GCSE students [86]. The students had access to AI tools for subjects like mathematics, English, science, and computer science, and were supported by three learning coaches. The students had access to personalized learning pathways that enabled them to revisit complex topics repeatedly and obtain feedback in real time. In this manner, students' knowledge gaps were identified and filled with deeper conceptual understanding [87]. David Game College offers a unique model where AI-human coaches complement each other to deliver targeted and personalized learning experiences. However, the costs per student for this program neared £27,000 per year, which is cited as a limitation to the program's scalability [87].

7. Insights and practical takeaways from global implementations

The case studies and literature presented in this chapter offer essential takeaways for educators and policymakers who seek to fully harness the potential of AI technology for personalized learning.

7.1 AI-supported personalization drives engagement and autonomy

A pattern that is unmistakable in personalized learning, across diverse contexts like China's Squirrel AI to the David Game College model, is that it boosts student motivation and engagement and results in tangible academic outcomes. This is because AI-supported learning platforms successfully align learning pathways with students' cognitive strengths and pace. The adaptive content delivery, combined with real-time feedback, empowers students with autonomy and improves confidence and well-being.

7.2 Scale is feasible with the right infrastructure

As seen in the case of China's Squirrel A, which is serving millions of students, it is possible to make scalable, personalized learning models. With robust digital infrastructure and data systems, such scalability is attainable but needs to be combined with relevant, expert-supported content recommendation and performance metric tools.

7.3 Human-AI synergy is essential

While AI provides unprecedented advantages in personalized learning, it is best harnessed when combined with experts' and educators' intervention, as seen in the hybrid learning model adopted by David Game College. AI tools must be complemented with human empathy, expert insights on pedagogy, and mentorship in education for the best outcomes for students.

7.4 Data is the engine of personalization

AI platforms like MATHia, Coursera's recommender engine, and others collect and collate large amounts of data for analysis, enabling them to deliver relevant recommendations for learning pathways and filling knowledge gaps. AI-supported personalization in learning, therefore, must prioritize investment in data systems that are interoperable and enable data collection at diverse points.

7.5 Training and capacity building are critical

There is a need for onboarding teachers not just on the technical operations but also to include a pedagogical understanding of how AI-supported personalization is derived. Educators need to learn to interpret data, adjust intervention strategies, and provide motivational support and guidance to students.

7.6 Ethics must guide AI deployment in education

Ethical concerns around issues of privacy and consent, discrimination and algorithm bias, and transparency need to be addressed and embedded in all stages of AI-supported personalization in education. Ethics should be made integral right from the stage of data collection to model design and instructional deployment. All AI-supported personalization needs to ensure the core values of equity, fairness, and respect for student agency.

8. Future trends: Virtual and augmented reality

The next frontier in AI-supported personalization lies in immersive technologies like virtual reality (VR), augmented reality (AR), or extended reality (XR).

8.1 Immersive and experiential learning

VR and AR enable immersive environments to allow students to engage in simulated real-world or abstract scenarios [88]. For example, VR can take students inside the human circulatory system or give them a glimpse of the ruins of ancient Rome, providing memorable experiential learning unmatched by videos or books [89]. AR overlays interactive content onto the physical world, which gives a sense of control to learners by allowing them to manipulate virtual models in space, enabling better retention and conceptual understanding [89, 90]. The VictoryXR platform is now available to over 100 colleges across the world and allows students to experiment with and explore 3D models [91]. zSpace is another tool that helps K-12 students virtually dissect frogs or explore science concepts in simulations using AR and zSpace [92].

8.2 AI + XR: Intelligent personalization in 3D worlds

Personalization of learning is taken to the next level with the convergence of AI and XR technologies. AI can personalize VR/AR experiences by adapting learning paths within immersive environments based on student performance and interaction data. AI-powered virtual chemistry labs that include XR can adjust the complexity of experiments to match the learners' level. Similarly, affective AI systems using facial expression analysis, gaze tracking, and motion data, which are used in VR, can offer accurate and relevant motivational or corrective interventions in real-time. An example is AI–VR tutors, which deliver content based on an analysis of the learners' responses and guide them on 3D tasks [93]. Furthermore, such AI–XR platforms can also enable collaborative learning with team-based challenges [94].

8.3 Future applications and challenges

XR technologies, comprising VR, AR, and mixed reality (MR), reshape the educational landscape and hold immense potential for vocational training, special education, and STEM education [95]. XR technologies can create safe, simulated environments for practicing high-risk procedures, such as surgeries or flight simulations. In special education, AR applications can provide social cues to children with autism [95]. Student engagement can be enhanced for engineering and technical education programs using XR modules [96]. These future applications may be challenging to implement due to high costs, limitations of bandwidth, lack of user training, access to devices and networks, and lack of scalability in adoption. XR is likely to be driven by the integration of AI, IoT (Internet of Things), and multimodal interfaces that enable personalized learning [97]. Nevertheless, policymakers, technology developers, and educators need to ensure equitable and ethical access for successful operations in immersive environments [96].

8.4 Research and innovation directions

There are several studies currently underway, like the one by Stanford University's Virtual Human Interaction Lab (VHIL), that are focused on understanding how VR shapes learning, empathy, and social understanding [97].

Similarly, UNESCO's Future of Learning initiative has started a global dialogue on ethical frameworks for XR technologies [96]. Some insights have emerged from pilot studies on frequent usage of VR environments by students of biology, which have found that VR environments lead to stronger engagement (1–1.5 times) and student motivation as compared to traditional learning over the longer term [94]. AR applications, too, have shown gains in memory retention ranging from 7–21% when compared to non-AR instruction [3, 94].

9. Summary

This chapter explored AI-enabled personalized learning in modern education. It introduced personalized learning and its historical background and outlined the five core objectives as: exploring the AI-supported personalization frameworks, identifying the key functions and tools used, exploring its implications, and showcasing implementation models leading to insights on success factors. The fundamental factors and theoretical underpinnings – constructivism, multiple intelligences, SDT, connectivism, and data-driven decision-making — and their contribution to AI-led personalized education systems at SPS, ViLLE, SLS Singapore, and others are discussed. AI tools are discussed as enablers of engagement, autonomy, and self-regulation through theory-based approaches.

The chapter also discusses key dimensions of AI-supported personalized learning in the form of learner engagement, motivation, learner-centric instructional design, data-driven real-time adaptive content delivery, and feedback supporting self-regulated learning. Further, ITS (e.g., MATHia, Korbit), recommender systems (e.g., Coursera), affective computing, and real-time feedback tools (e.g., Amira, Eedi) are discussed to demonstrate how they lead to measurable gains in academic performance and student well-being.

SPS, Carnegie Learning, Squirrel AI in China, and David Game College in the UK are used as case studies to illustrate scalable deployment, provide insights, and takeaways. The key learnings are identified as follows: AI improves engagement, it can be deployed at scale, data is the core aspect of AI-led personalization, human-AI synergy is essential, and ethics must guide deployment. Future trends, particularly AI-powered XR (VR and AR) in STEM, vocational training, and special education, are also discussed.

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